

components of an LLM can be deployed across multiple containers, potentially on different nodes in a cluster. This not only enhances scalability but also enables fault-tolerant and resilient AI systems.

The intersection of generative AI and Docker also holds significant potential for advancing the integration of machine learning and DevOps (MLOps) practices. MLOps, which focuses on the continuous delivery and automation of ML models, can greatly benefit from Docker's capabilities in environment standardisation and dependency management. Research can delve into creating streamlined pipelines where LLMs are trained, tested, and deployed within Docker containers,

ensuring consistency across all stages of the machine learning lifecycle. Furthermore, the immutable nature of Docker containers can help in maintaining version control and traceability of models, which is crucial for reproducibility and auditability in AI research.

Another critical area for research is the security implications of deploying generative AI models within Docker containers. As these models become increasingly integrated into business-critical applications, ensuring their security against vulnerabilities and

attacks becomes paramount. Docker's security features, such as container isolation and image signing, can be leveraged to enhance the security posture of LLM deployments. However, the complexity of AI models also introduces new vectors of attack, such as model poisoning and adversarial inputs, which require sophisticated defence mechanisms. Researchers can investigate the development of robust security protocols and compliance frameworks tailored to the unique challenges posed by AI-container integration. **END** 🐧

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Kubernetes. These platforms provide the tools needed to create, manage, and orchestrate containers.

Containerise applications:

Encapsulate your applications and their dependencies into containers. This involves creating a Docker file that specifies the application's environment and dependencies.

Use Infrastructure as Code (IaC):

Tools like Terraform and Ansible can be used to define and manage infrastructure across multiple clouds.

Container orchestration: Use Kubernetes to manage containers and applications across different environments.

Configuration management:

Automate configuration tasks and ensure consistency across different systems using tools like Ansible or Puppet.

Networking: Implement network virtualisation and management using tools like Open vSwitch (OVN) and Calico.

Data management: Use tools like Apache Hadoop and Apache Spark for Big Data processing and analytics.

Security: Protect applications and data in multi-cloud environments

using open source security tools like OpenSCAP and OWASP ZAP.

Containerisation vs virtualisation

The technology stack is similar for containerisation and virtualisation. Hence, the deployment model is largely decided based on the needs of the application's non-functional characteristics. One way to decide is by making a checklist of the benefits

and concerns of deploying each model. Table 3 outlines the pros and cons of both virtualisation and containerisation.

Open source containerisation software, a cornerstone of microservices driven architecture, is highly scalable and performs well at a fraction of the traditional or virtualisation resource costs. But it does require careful planning of architecture and design to address management and security concerns. **END** 🐧

📧 References

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